

# Лабораторная работа № 9

## Настройка IMAP-сервера

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### Цель:

Получить навыки установки и настройки Dovecot для доступа к почте по IMAP.



# Установка Dovecot

```
root@server:~  
root@server ~]# dnf -y install dovecot telnet  
Last metadata expiration check: 0:22:32 ago on Sun 06 Sep 2026 08:51:38 AM UTC.  
Dependencies resolved.  
=====
```

| Package     | Architecture | Version              | Repository | Size  |
|-------------|--------------|----------------------|------------|-------|
| Installing: |              |                      |            |       |
| dovecot     | x86_64       | 1:2.3.21-19.el10_2.1 | appstream  | 4.8 M |
| telnet      | x86_64       | 1:0.17-94.el10       | appstream  | 62 k  |

```
=====
```

Transaction Summary

Install 2 Packages

Total download size: 4.8 M  
Installed size: 18 M  
Downloading Packages:

| (1/2): telnet-0.17-94.el10.x86_64.rpm        | 306 kB/s | 62 kB  | 00:00 |
|----------------------------------------------|----------|--------|-------|
| (2/2): dovecot-2.3.21-19.el10_2.1.x86_64.rpm | 7.2 MB/s | 4.8 MB | 00:00 |
| -----                                        |          |        |       |
| Total                                        | 4.7 MB/s | 4.8 MB | 00:01 |

```
-----
```

Running transaction check  
Transaction check succeeded.  
Running transaction test  
Transaction test succeeded.  
Running transaction  
 Preparing : 1/1  
 Installing : telnet-1:0.17-94.el10.x86\_64 1/2  
 Running scriptlet: dovecot-1:2.3.21-19.el10\_2.1.x86\_64 2/2  
 Installing : dovecot-1:2.3.21-19.el10\_2.1.x86\_64 2/2  
 Running scriptlet: dovecot-1:2.3.21-19.el10\_2.1.x86\_64 2/2  
 Complete!

Installed:

dovecot-1:2.3.21-19.el10\_2.1.x86\_64 telnet-1:0.17-94.el10.x86\_64

```
complete!  
root@server ~]# nano /etc/dovecot/dovecot.conf  
root@server ~]# nano /etc/dovecot/conf.d/10-auth.conf  
root@server ~]# nano /etc/dovecot/conf.d/auth-system.conf.ext  
root@server ~]# nano /etc/dovecot/conf.d/10-mail.conf  
root@server ~]# grep '^protocols' /etc/dovecot/dovecot.conf  
protocols = imap pop3  
root@server ~]# grep '^auth_mechanisms' /etc/dovecot/conf.d/10-auth.conf  
auth_mechanisms = plain  
root@server ~]# grep '^mail_location' /etc/dovecot/conf.d/10-mail.conf  
mail_location = maildir:~/Maildir  
root@server ~]#
```

## Смысл результата

На сервер установлен Dovecot, включены IMAP/POP3-службы и проверен статус процесса. Firewalld разрешает необходимые почтовые сервисы.

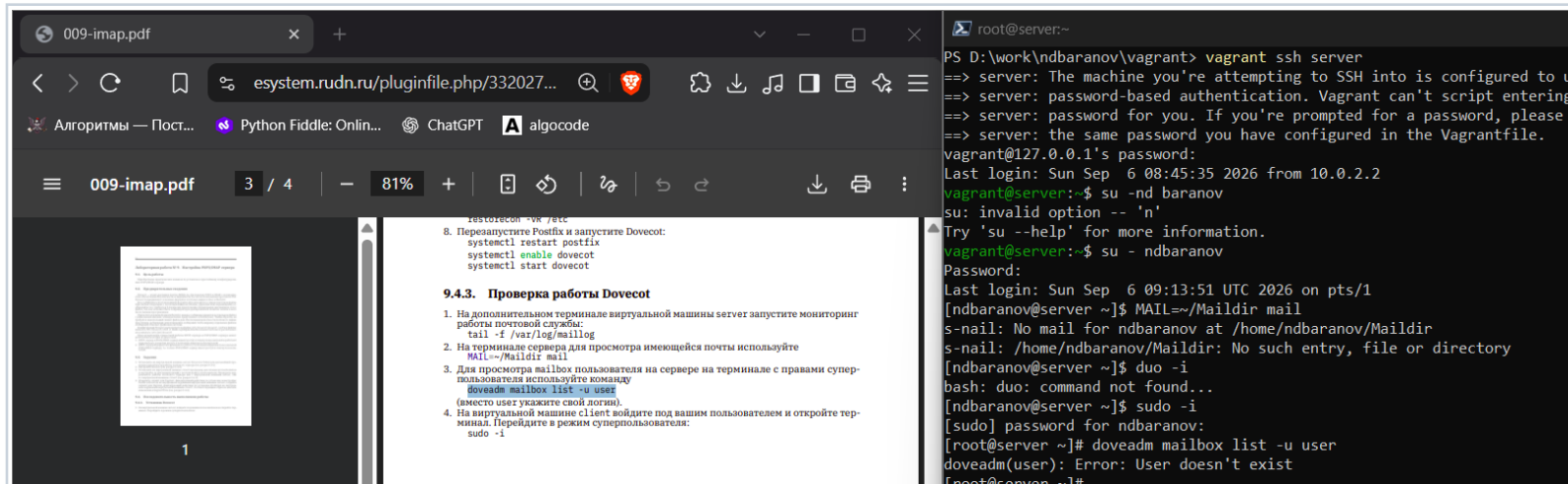
# Параметры протоколов

```
root@server:~# postconf -e 'home_mailbox = Maildir/'
root@server ~]# firewall-cmd --get-services
firewall-cmd --add-service=pop3 --permanent
firewall-cmd --add-service=pop3s --permanent
firewall-cmd --add-service=imap --permanent
firewall-cmd --add-service=imaps --permanent
firewall-cmd --reload
firewall-cmd --list-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls d
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mnpd mon
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi
os-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag
econsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netdrv ptp pulseaudio puppetmaster quassel radi
us radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client
samba-de sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-t
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsd vnc-server vrrp warpinator
wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di
scovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
success
success
success
success
success
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
root@server ~]# restorecon -vR /etc
root@server ~]# systemctl restart postfix
systemctl enable dovecot
systemctl start dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' → '/usr/lib/systemd/system
/dovecot.service'.
root@server ~]# systemctl status dovecot --no-pager
* dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-06 09:19:27 UTC; 9s ago
   Invocation: 066edb71ab784c7ab4df243997df15ae
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
   Main PID: 15762 (dovecot)
   Status: "v2.3.21 (47349e2482) running"
```

## Смысл результата

В конфигурации включены imap и pop3, настроены механизмы обычной системной аутентификации. dovecot выводит фактически применённые значения.

# Переход на Maildir



## Смысл результата

Параметр `mail_location` указывает на `~/Maildir`. В отличие от единого `mbox` каждое письмо хранится отдельным файлом.

# Создание структуры Maildir

```
root@server:~  
PS D:\work\ndbaranov\vagrant> vagrant ssh server  
==> server: The machine you're attempting to SSH into is configured to use  
==> server: password-based authentication. Vagrant can't script entering the  
==> server: password for you. If you're prompted for a password, please enter  
==> server: the same password you have configured in the Vagrantfile.  
vagrant@127.0.0.1's password:  
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2  
vagrant@server:~$ su -nd baranov  
su: invalid option -- 'n'  
Try 'su --help' for more information.  
vagrant@server:~$ su - ndbaranov  
Password:  
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1  
[ndbaranov@server ~]$ MAIL=~/.Maildir mail  
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir  
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory  
[ndbaranov@server ~]$ duo -i  
bash: duo: command not found...  
[ndbaranov@server ~]$ sudo -i  
[sudo] password for ndbaranov:  
[root@server ~]# dovecadm mailbox list -u user  
doveadm(user): Error: User doesn't exist  
[root@server ~]# dovecadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]#
```

## Смысл результата

Для пользователя созданы каталоги cur, new и tmp. Новое письмо появляется в new, что подтверждает правильное место доставки.

```
root@client:~
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2
vagrant@server:~$ su -nd baranov
su: invalid option -- 'n'
Try 'su --help' for more information.
vagrant@server:~$ su - ndbaranov
Password:
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1
[ndbaranov@server ~]$ MAIL=~/.Maildir mail
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory
[ndbaranov@server ~]$ duo -i
bash: duo: command not found...
[ndbaranov@server ~]$ sudo -i
[sudo] password for ndbaranov:
[root@server ~]# doveadm mailbox list -u user
doveadm(user): Error: User doesn't exist
[root@server ~]# doveadm mailbox list -u ndbaranov
INBOX
[root@server ~]# exit
logout
[ndbaranov@server ~]$ exit
logout
vagrant@server:~$ exit
logout
PS D:\work\ndbaranov\vagrant> vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:50:22 2026 from 10.0.2.2
vagrant@client:~$ sudo -i
root@client:~# dnf -y install evolution
Last metadata expiration check: 0:12:50 ago on Sun 06 Sep 2026 09:09:47 AM UTC.
Dependencies resolved.
=====
Package                Architecture Version                Repository    Size
=====
Installing:
evolution               x86_64        3.52.4-2.el10_1       epel          4.6 M
Installing dependencies:
cmack-lib               x86_64        0.30.3-5.el10_0       epel          96 k
cpuinfo                 x86_64        24.09.26-1.git1e83a2f.el10_1.1 epel          42 k
evolution-langpacks     noarch        3.52.4-2.el10_1       epel          6.0 M
flite                   x86_64        2.2-10.el10_0         epel          12 M
gspell                  x86_64        1.14.0-1.el10_0       epel          111 k
harfbuzz-icu            x86_64        8.4.0-6.el10          appstream     15 k
highlight               x86_64        4.20-3.el10_2         epel          951 k
highway                 x86_64        1.3.0-1.el10_2        epel          505 k
=====
```

## Смысл результата

Через telnet установлено соединение с портом 143 и выполнена команда LOGIN. Ответ ОК показывает успешную аутентификацию пользователя.

# Работа с папкой INBOX

```
root@client:~  
root@client:~# id ndbaranov  
id: 'ndbaranov': no such user  
root@client:~# useradd -m ndbaranov  
echo 'ndbaranov:123456' | chpasswd  
root@client:~# postconf -e 'inet_protocols = ipv4'  
systemctl enable --now postfix  
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.  
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet  
Connecting to 192.168.1.1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify return:1  
*. OK Pre-login capabilities listed, post-login capabilities have more.  
a1  
a1 BAD Error in IMAP command received by server.  
LOGIN  
LOGIN BAD Error in IMAP command received by server.  
ndbaranov  
* BYE Too many invalid IMAP commands.  
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet  
Connecting to 192.168.1.1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify return:1  
*. OK Pre-login capabilities listed, post-login capabilities have more.  
a1 LOGIN ndbaranov 123456  
* CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS THREAD=ORDEREDSUBJECT MULTIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-EXTENDED I18NLEVEL=1 CONDSTORE QRESYNC ESEARCH ESORT SEARCHRES WITHIN CONTEXT=SEARCH LIST-STATUS BINARY MOVE SNIPPET=FUZZY PREVIEW=FUZZY PREVIEW STATUS=SIZE SAVEDATE LITERAL+ NOTIFY SPECIAL-USE  
a1 OK Logged in  
a2 LIST "" ""  
* LIST (\HasNoChildren) "." INBOX  
a2 OK List completed (0.001 + 0.000 secs).  
a3 SELECT INBOX  
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)  
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.  
* 0 EXISTS  
* 0 RECENT  
* OK [UIDVALIDITY 1788688840] UIDs valid  
* OK [UIDNEXT 1] Predicted next UID  
a3 OK [READ-WRITE] Select completed (0.039 + 0.000 + 0.038 secs).  
a4 FETCH 1:* BODY[HEADER.FIELDS (FROM TO SUBJECT)]  
a4 BAD Error in IMAP command FETCH: Invalid messageset (0.001 + 0.000 secs).  
a5 LOGOUT  
* BYE Logging out  
a5 OK Logout completed (0.001 + 0.000 secs).  
root@client:~#
```

## Смысл результата

Команды LIST, SELECT и FETCH вывели папки и содержимое письма. Сервер корректно обрабатывает основные операции IMAP.

# Удаление письма через IMAP

```
ndbaranov@server:~
[root@server ~]# doveadm mailbox list -u ndbaranov
INBOX
[root@server ~]# doveadm search -u ndbaranov mailbox INBOX ALL
ada75421c8399d6a63400000a2dd03fa 1
ada75421c8399d6a63400000a2dd03fa 2
ada75421c8399d6a63400000a2dd03fa 3
[root@server ~]# doveadm mailbox list -u ndbaranov
INBOX
[root@server ~]# su - ndbaranov
Last login: Sun Sep  6 09:20:29 UTC 2026 on pts/2
[ndbaranov@server ~]$ MAIL=~/.Maildir mail
s-nail version v14.9.24. Type '?' for help
/home/ndbaranov/Maildir: 3 messages 3 new
+N 1 Super User      2026-09-06 10:02 17/664 "IMAP test 1
  N 2 Super User      2026-09-06 10:02 17/664 "IMAP test 2
  N 3 Super User      2026-09-06 10:02 17/666 "IMAP test 3
& q
Held 3 messages in /home/ndbaranov/Maildir
[ndbaranov@server ~]$ telnet mail.ndbaranov.net 110
Trying 192.168.1.1...
Connected to mail.ndbaranov.net.
Escape character is '^]'.
+OK Dovecot ready.
user ndbaranov
+OK
pass 123456
+OK Logged in.
list
+OK 3 messages:
1 679
2 679
3 681
.
retr 1
+OK 679 octets
Return-Path: <root@client.ndbaranov.net>
X-Original-To: ndbaranov@ndbaranov.net
Delivered-To: ndbaranov@ndbaranov.net
Received: from client.ndbaranov.net (unknown [192.168.1.31])
        by server.ndbaranov.net (Postfix) with ESMTPS id E051B6007316
        for <ndbaranov@ndbaranov.net>; Sun,  6 Sep 2026 10:02:12 +0000 (UTC)
Received: by client.ndbaranov.net (Postfix, from userid 0)
        id 9CA0220E1589; Sun,  6 Sep 2026 10:02:12 +0000 (UTC)
Date: Sun, 06 Sep 2026 10:02:12 +0000
To: ndbaranov@ndbaranov.net
Subject: IMAP test 1
User-Agent: s-nail v14.9.24
Message-Id: <20260906100212.9CA0220E1589@client.ndbaranov.net>
From: Super User <root@client.ndbaranov.net>

message one
.
dele 2
+OK Marked to be deleted.
quit
```

## Смысл результата

Флаг Deleted назначен командой STORE, а EXPUNGE окончательно удалил сообщение. Повторная проверка показывает пустую папку.

## Результат

Dovecot хранит сообщения в Maildir и предоставляет доступ по IMAP. Проверены аутентификация, просмотр папок, чтение, установка флагов и удаление письма.